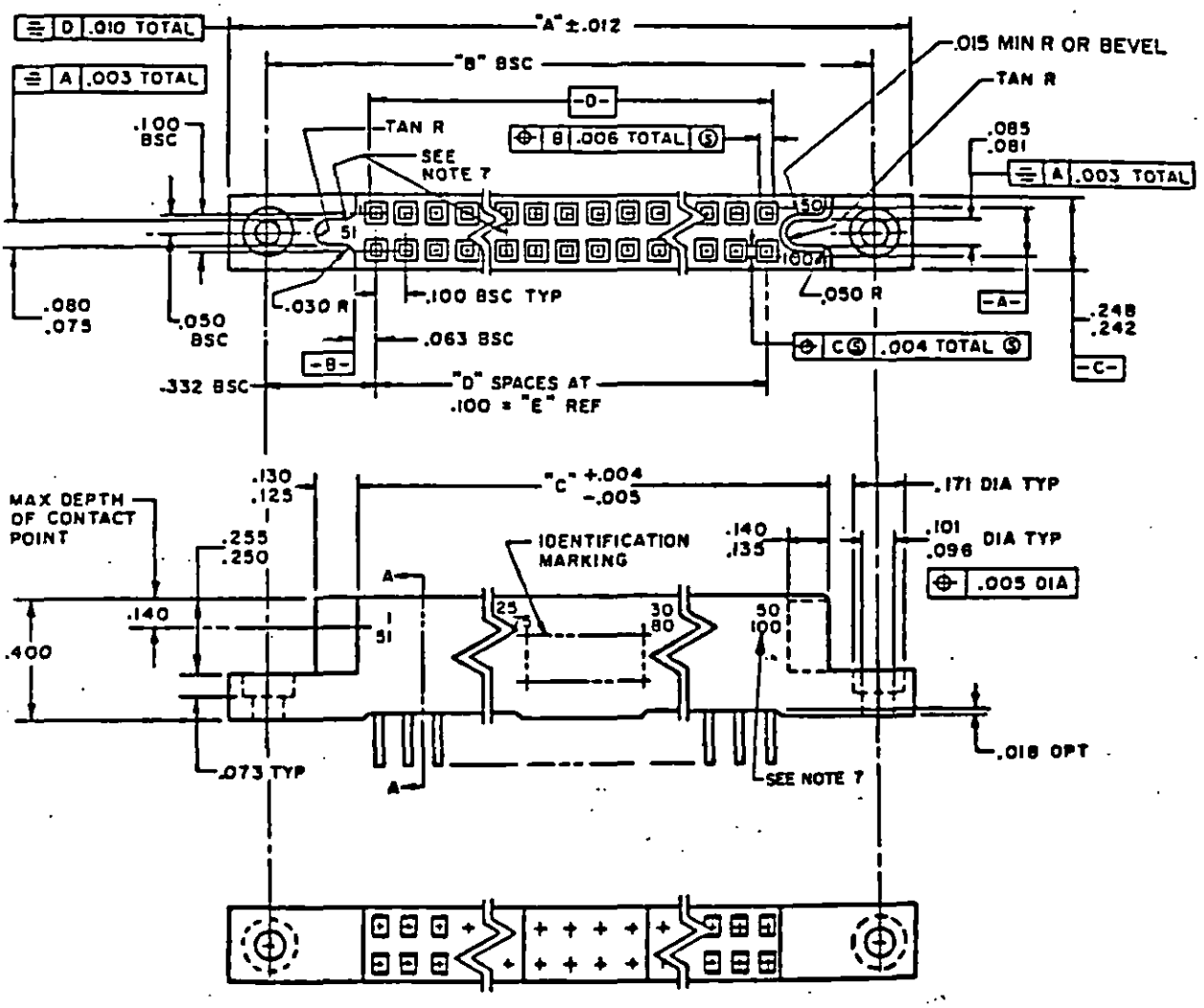
MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES: RECEPTACLES 30 THRU 140 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.100 SQ. GRID)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the connectors described herein shall consist of this specification and the latest issue of MIL-C-55302.



Style 1 (internal keying).

FIGURE 1. Connectors, receptacle (.100 square grid) - Continued.

Ⓖ denotes changes

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1/ Dash number	Number contacts	Type	A	B	C	D	E	Post length type	Contact identification numbers
25	30	I							
26	30	II						Not applicable	
27	30	III							
28	30	IV							
37	30	V	2.355 (59.82)	2.065 (52.45)	1.665 (42.29)	14	1.400 (35.56)	1	1 - 15
38	30	VI						1	16 - 30
39	30	V						2	
40	30	VI						2	
41	30	V						3	
42	30	VI						3	
29	40	I							
30	40	II						Not applicable	
31	40	III							
32	40	IV							
43	40	V	2.855 (72.52)	2.565 (65.15)	2.165 (54.99)	19	1.900 (48.26)	1	1 - 20
44	40	VI						1	21 - 40
45	40	V						2	
46	40	VI						2	
47	40	V						3	
48	40	VI						3	
01	50	I							
02	50	II						Not applicable	
03	50	III							
04	50	IV							
49	50	V	3.355 (85.22)	3.065 (77.85)	2.665 (67.69)	24	2.400 (60.96)	1	1 - 25
50	50	VI						1	26 - 50
51	50	V						2	
52	50	VI						2	
53	50	V						3	
54	50	VI						3	
05	60	I							
06	60	II						Not applicable	
07	60	III							
08	60	IV							
55	60	V	3.855 (97.92)	3.565 (90.55)	3.165 (80.39)	29	2.900 (73.66)	1	1 - 30
56	60	VI						1	31 - 60
57	60	V						2	
58	60	VI						2	
59	60	V						3	
60	60	VI						3	

1/ See note at end of table.

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

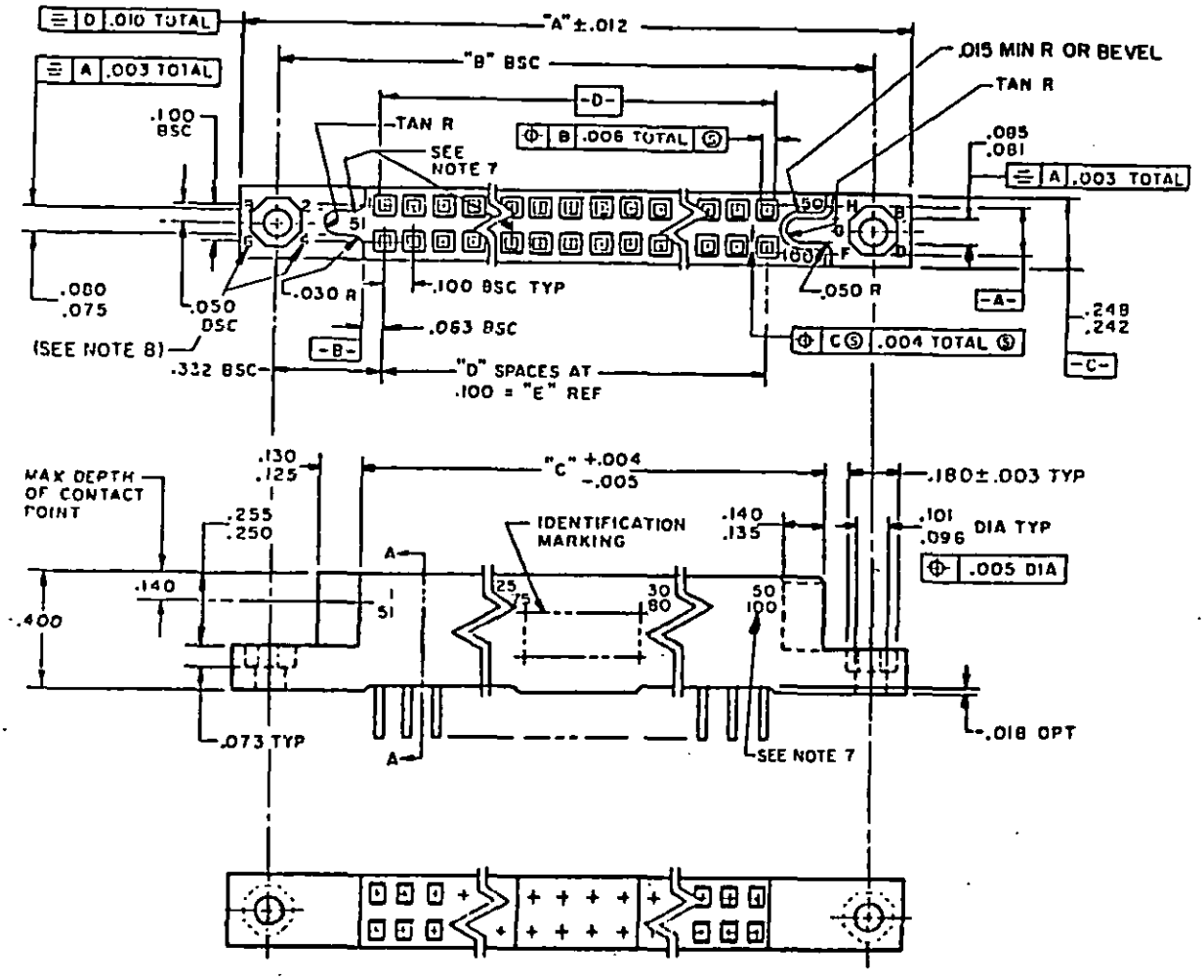
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1/ Dash number	Number contacts	Type	A	B	C	D	E	Post length type	Contact identification numbers
33	70	I							
34	70	II						Not applicable	
35	70	III							
36	70	IV							
61	70	V	4.355 (110.62)	4.065 (103.25)	3.665 (93.09)	34	3.400 (86.36)	1	1 - 35
62	70	VI						1	36 - 70
63	70	V						2	
64	70	VI						2	
65	70	V						3	
66	70	VI						3	
09	80	I							
10	80	II						Not applicable	
11	80	III							
12	80	IV							
67	80	V	4.855 (123.32)	4.565 (115.95)	4.165 (105.79)	39	3.900 (99.06)	1	1 - 40
68	80	VI						1	41 - 80
69	80	V						2	
70	80	VI						2	
71	80	V						3	
72	80	VI						3	
13	90	I							
14	90	II						Not applicable	
15	90	III							
16	90	IV							
73	90	V	5.355 (136.02)	5.065 (128.65)	4.665 (118.49)	44	4.400 (111.76)	1	1 - 45
74	90	VI						1	46 - 90
75	90	V						2	
76	90	VI						2	
77	90	V						3	
78	90	VI						3	
17	100	I							
18	100	II						Not applicable	
19	100	III							
20	100	IV							
79	100	V	5.855 (148.72)	5.565 (141.35)	5.165 (131.19)	49	4.900 (124.46)	1	1 - 50
80	100	VI						1	51 - 100
81	100	V						2	
82	100	VI						2	
83	100	V						3	
84	100	VI						3	
21	110	I							
22	110	II						Not applicable	
23	110	III							
24	110	IV							
85	110	V	6.355 (161.42)	6.065 (154.05)	5.665 (143.89)	54	5.400 (137.16)	1	1 - 55
86	110	VI						1	56 - 110
87	110	V						2	
88	110	VI						2	
89	110	V						3	
90	110	VI						3	

1/ Dash numbers 01 through 90 are for style 1 (internal keying) connectors.

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

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Style 2 (external keying).

FIGURE 1. Connectors, receptacles, (.100 square grid) - Continued.

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Dash 1/ number	Number contacts	Type	A	B	C	D	E	Post length type	Contact identification numbers
91	30	I						N/A	
100	30	II						N/A	
109	30	III						N/A	
118	30	VI						3	
127	30	VI	2.355 (59.82)	2.065 (52.45)	1.665 (42.29)	14	1.400 (35.56)	2	1 - 15
136	30	VI						1	16 - 30
145	30	V						3	
154	30	V						2	
163	30	V						1	
172	30	IV						N/A	
92	40	I						N/A	
101	40	II						N/A	
110	40	III						N/A	
119	40	VI						3	
128	40	VI	2.855 (72.52)	2.565 (65.15)	2.165 (54.99)	19	1.900 (48.26)	2	1 - 20
137	40	VI						1	21 - 40
146	40	V						3	
155	40	V						2	
164	40	V						1	
173	40	IV						N/A	
93	50	I						N/A	
102	50	II						N/A	
111	50	III						N/A	
120	50	VI	3.355 (85.22)	3.065 (77.85)	2.665 (67.69)	24	2.400 (60.96)	3	1 - 25
129	50	VI						2	26 - 50
138	50	VI						1	
147	50	V						3	
156	50	V						2	
165	50	V						1	
174	50	IV						N/A	
94	60	I						N/A	
103	60	II						N/A	
112	60	III						N/A	
121	60	VI	3.855 (97.92)	3.565 (90.55)	3.165 (80.39)	29	2.900 (73.66)	3	1 - 30
130	60	VI						2	31 - 60
139	60	VI						1	
148	60	V						3	
157	60	V						2	
166	60	V						1	
175	60	IV						N/A	
95	70	I						N/A	
104	70	II						N/A	
113	70	III						N/A	
122	70	VI	4.355 (110.62)	4.065 (103.25)	3.665 (93.09)	34	3.400 (86.36)	3	1 - 35
131	70	VI						2	36 - 70
140	70	VI						1	
149	70	V						3	
158	70	V						2	
167	70	V						1	
176	70	IV						N/A	

1/ See note at end of table.

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

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1/ Dash number	Number contacts	Type	A	B	C	D	E	Post length type	Contact identification numbers
96	80	I						N/A	
105	80	II						N/A	
114	80	III						N/A	
123	80	VI	4.855 (123.32)	4.565 (115.95)	4.165 (105.79)	39	3.900 (99.06)	3	1 - 40
132	80	VI						2	41 - 80
141	80	VI						1	
150	80	V						3	
159	80	V						2	
168	80	V						1	
177	80	IV						N/A	
97	90	I						N/A	
106	90	II						N/A	
115	90	III						N/A	
124	90	VI	5.355 (136.02)	5.065 (128.65)	4.665 (118.49)	44	4.400 (111.76)	3	1 - 45
133	90	VI						2	46 - 90
142	90	VI						1	
151	90	V						3	
160	90	V						2	
169	90	V						1	
178	90	IV						N/A	
98	100	I						N/A	
107	100	II						N/A	
116	100	III						N/A	
125	100	VI	5.855 (148.72)	5.565 (141.35)	5.165 (131.19)	49	4.900 (124.46)	3	1 - 50
134	100	VI						2	51 - 100
143	100	VI						1	
152	100	V						3	
161	100	V						2	
170	100	V						1	
179	100	IV						N/A	
99	110	I						N/A	
108	110	II						N/A	
117	110	III						N/A	
126	110	VI	6.355 (161.42)	6.065 (154.05)	5.665 (143.89)	54	5.400 (137.16)	3	1 - 55
135	110	VI						2	56 - 110
144	110	VI						1	
153	110	V						3	
162	110	V						2	
171	110	V						1	
180	110	IV						N/A	

1/ See note at end of table.

FIGURE 1. Connectors, receptacles, (.100 square grid) - Continued.

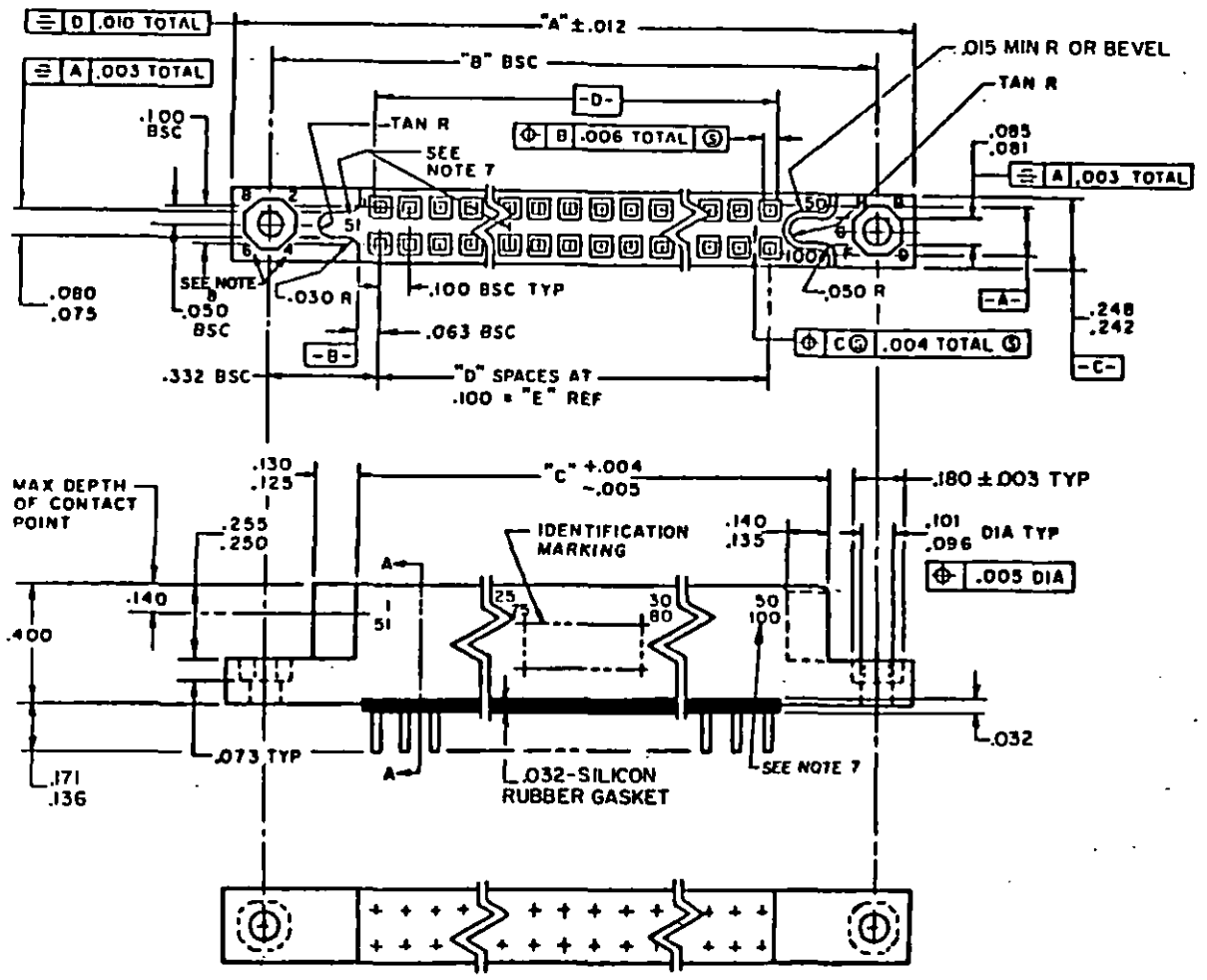
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Dash 1/ number	Number contacts	Type	A	B	C	D	E	Post length type	Contact identification numbers
190	120	I						N/A	
194	120	II						N/A	
198	120	III						N/A	
202	120	VI						3	
206	120	VI	6.855	6.565	6.165	59	5.900	2	1-60
210	120	VI	(174.18)	(166.75)	(156.60)		(149.86)	1	61-120
214	120	V						3	
218	120	V						2	
222	120	V						1	
226	120	IV						N/A	
191	130	I						N/A	
195	130	II						N/A	
199	130	III						N/A	
203	130	VI						3	
207	130	VI	7.355	7.065	6.665	64	6.400	2	1-65
211	130	VI	(186.82)	(179.45)	(169.30)		(162.56)	1	66-130
215	130	V						3	
219	130	V						2	
223	130	V						1	
227	130	IV						N/A	
192	134	I						N/A	
196	134	II						N/A	
200	134	III						N/A	
204	134	VI						3	
208	134	VI						2	
212	134	VI	7.555	7.265	6.865	66	6.600	1	1-67
216	134	V	(191.90)	(184.53)	(174.38)		(167.64)	3	68-134
220	134	V						2	
224	134	V						1	
228	134	IV						N/A	
193	140	I						N/A	
197	140	II						N/A	
201	140	III						N/A	
205	140	VI						3	
209	140	VI	7.855	7.565	7.165	69	6.900	2	1-70
213	140	VI	(199.52)	(192.15)	(181.99)		(175.26)	1	71-140
217	140	V						3	
221	140	V						2	
225	140	V						1	
229	140	IV						N/A	

1/ Dash numbers 91 through 180 and 190 through 229 are for style 2 (external keying) connectors.

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

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Style 3 (silicon rubber gasket).

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

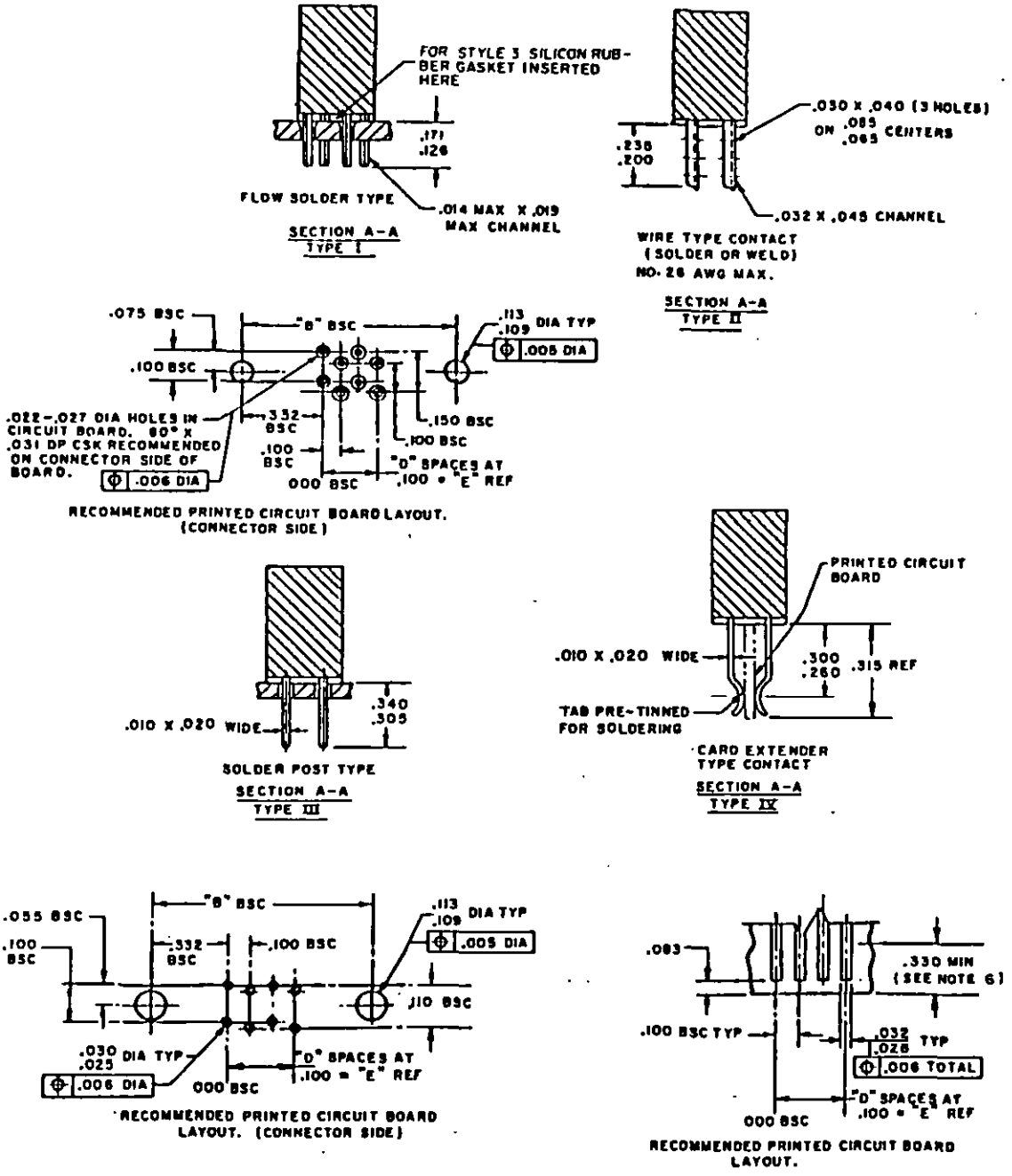
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1/ Dash number	Number contacts	Type	A	B	C	D	E	Contact identification numbers
181	30	I	2.355 (59.82)	2.065 (52.45)	1.665 (42.29)	14	1.400 (35.56)	1-15 16-30
182	40	I	2.855 (72.52)	2.565 (65.15)	2.165 (54.99)	19	1.900 (48.26)	1-20 21-40
183	50	I	3.355 (85.22)	3.065 (77.85)	2.665 (67.69)	24	2.400 (60.96)	1-25 26-50
184	60	I	3.855 (97.92)	3.565 (90.55)	3.165 (80.39)	29	2.900 (73.66)	1-30 31-60
185	70	I	4.355 (110.62)	4.065 (103.25)	3.665 (93.09)	34	3.400 (86.36)	1-35 36-70
186	80	I	4.855 (123.32)	4.565 (115.95)	4.165 (105.79)	39	3.900 (99.06)	1-40 41-80
187	90	I	5.355 (136.02)	5.065 (128.65)	4.665 (118.49)	44	4.400 (111.76)	1-45 46-90
188	100	I	5.855 (148.72)	5.565 (141.35)	5.165 (131.19)	49	4.900 (124.46)	1-50 51-100
189	110	I	6.355 (161.42)	6.065 (154.05)	5.665 (143.89)	54	5.400 (137.16)	1-55 56-110

1/ Dash numbers 181 through 189 are for style 03 (silicon rubber gasket) connectors.

FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

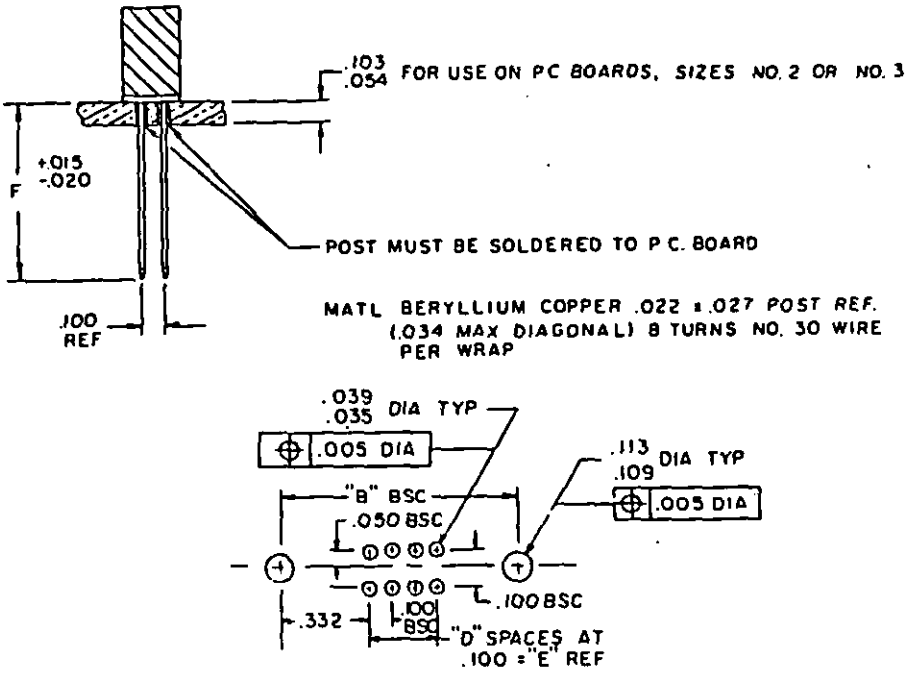
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Contact types and layouts

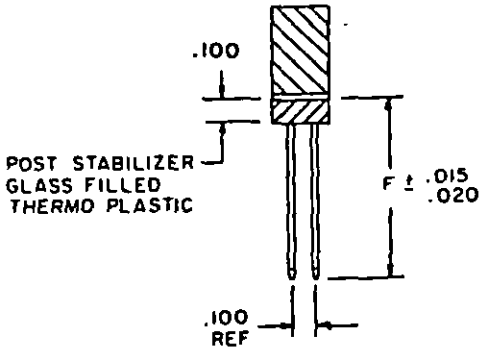
FIGURE 1. Connectors, receptacles,(.100 square grid) - Continued.

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RECOMMENDED P.C. BOARD LAYOUT
(CONNECTOR SIDE)

BOARD MOUNT TYPE V



RACK MOUNT TYPE VI

WIRE WRAPPED TYPE CONTACT
SECTION A-A

Post length type	F
1	.405 (10.29 mm)
2	.565 (14.35 mm)
3	.725 (18.42 mm)

Contact types and layouts

FIGURE 1. Connectors, receptacles.(.100 square grid) - Continued.

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Inches	mm	Inches	mm	Inches	mm
.003	.08	.045	1.14	.130	3.30
.004	.10	.050	1.27	.135	3.43
.005	.13	.054	1.37	.136	3.45
.006	.15	.055	1.40	.140	3.56
.010	.25	.063	1.60	.150	3.81
.012	.30	.065	1.65	.171	4.34
.014	.36	.073	1.85	.180	4.57
.015	.38	.075	1.90	.200	5.08
.018	.46	.080	2.03	.235	5.97
.019	.48	.081	2.06	.242	6.15
.020	.51	.083	2.11	.248	6.30
.022	.56	.085	2.16	.250	6.35
.025	.64	.096	2.44	.255	6.48
.027	.69	.100	2.54	.260	6.60
.028	.71	.101	2.57	.300	7.62
.030	.76	.103	2.62	.305	7.75
.032	.81	.109	2.77	.315	8.00
.034	.86	.110	2.79	.330	8.38
.035	.89	.113	2.87	.332	8.43
.039	.99	.125	3.18	.340	8.64
.040	1.02	.126	3.20	.400	10.16

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm) on decimals and ±2° on angles.
4. These connectors mate with connectors specified in MIL-C-55302/26 and /110.
5. Use printed circuit boards with the type III receptacle and type V receptacle.
6. Minimum pad length on printed circuit board.
7. Numbers indicating, every 5 cavities stamped on side. Numbers indicating end cavities and lines indicating every 5 cavities stamped on mating face.
8. Location indicators embossed on surface.

FIGURE 1. Connectors, receptacles, (.100 square grid) - Continued.

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REQUIREMENTS

Design and construction:

Dimensions and configuration: See figures 1 and 2, and table I.

Material:
Gasket: Silicon rubber.
Ⓖ Plating: Engagement area: Gold per MIL-C-55302 for .100 length, over nickel all over per MIL-C-55302.

Termination area: Tin lead over nickel per MIL-C-55302 for the following lengths:

.165 minimum length for type I.
.200 minimum length for type II.
.300 minimum length for type III.
.140 minimum length for type IV.

Termination for type V and VI contacts:
Gold over nickel pre MIL-C-55302 for the entire termination length.

Contact identification: See figures 1 and 2.

Contact rating: 3.0 amperes maximum per contact, 2.25 amperes continuous per contact at 75°F.

Keying: See MIL-C-55302/31.

Oversized pin exclusion (sockets only): .0465 diameter pin.

Contact separation force: The connector shall meet the requirements of MIL-C-55302/29.

Wire size: Number 26 AWG maximum with type II, solder eyelet contact for wire termination. Number 30 AWG may be used with type V and type VI wrappost contacts.

Resistance to test probe damage: The connector shall meet the requirements of MIL-C-55302/30.

Mating and unmating: The maximum mating force in pounds shall be the number of contacts multiplied by .25 and the withdrawal force in pounds shall be a minimum of .025 times the number of contacts and shall not exceed the measured insertion force.

Contact resistance: No individual contact pair shall have a resistance exceeding .020 ohm.

Contact retention: 3 pounds minimum.

Dielectric withstanding voltage:
Sea level: 900 volts rms.
High altitude: 200 volts rms.

Replacement contacts: See MIL-C-55302/32.

Ⓖ Contact removal tool: See MIL-I-81969/9-01 for type I through type V. Use AMP tool 1-265871-7 for type VI.

Part number: M55302/27- (dash number from figure 1).

Supersession data: Style 2, type I supersedes type VII.

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government.

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Custodians:
Army - CR
Navy - EC
Air Force - 17

Review activities:
Army - CR, AR, MI
Navy - SH, AS, OS
Air Force - 11, 99
DLA - ES

User activities:
Army - ME, AT, AV
Navy - MC
Air Force - 19

Preparing activity:
Army - CR

Agent:
DLA - ES

(Project 5935-3645-05)

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